

Design Thinking in IT – Assignment 1

NA08

Introduction

This project focuses on designing a simpler and more user-friendly website (WattBuddy) that helps users better manage their energy usage. Due to increasing concerns about climate change and rising energy costs, it is becoming more significant for individuals to monitor their resource consumption. However, there are many existing tools often too complex or difficult to navigate, and it will discourage users from actively engaging in sustainable practices.

The aim of the project is to create a platform that allows users to track their usage, access useful tips, and set goals to better manage their energy consumption without feeling overwhelmed or distressed.

This project is based on design thinking process, including empathising, defining, ideating, prototyping, and testing. These steps help make sure the final product is user-friendly, reliable and correctly aligns with user needs.

Use of Design Thinking

Empathise

This is the first phase in Design thinking. It focuses on observing and understanding users to uncover deep and meaningful needs, their behaviours, and challenges they encounter in managing household energy consumption. For this project, the goal is to immerse ourselves in the users' shoes and experience it all.

Target users

After careful consideration, the following list of target users is developed:

- Youngster/Students living in shared accommodation
- Users with less tech knowledge
- Working individuals looking for simple and effective solutions to save energy
- Families trying to save money on bills

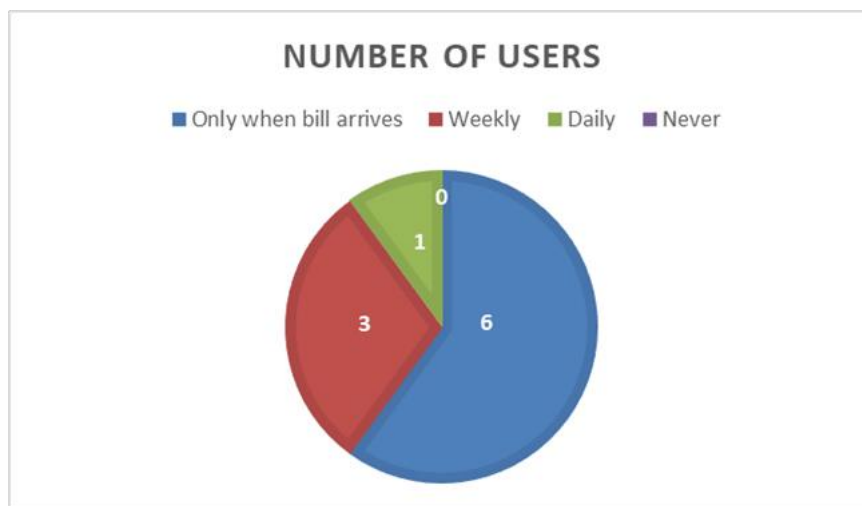
Since our website is meant to be simple and accessible that offers quick solutions, these users are likely to benefit the most.

User Research

For this part, a list of considerate questions was created carefully, and a survey was conducted. A certain number of people (10) were all asked the same questions and their responses were noted. These questions range from users' current personal experiences to more critical questions that would help with the website's design process. The results are as follows:

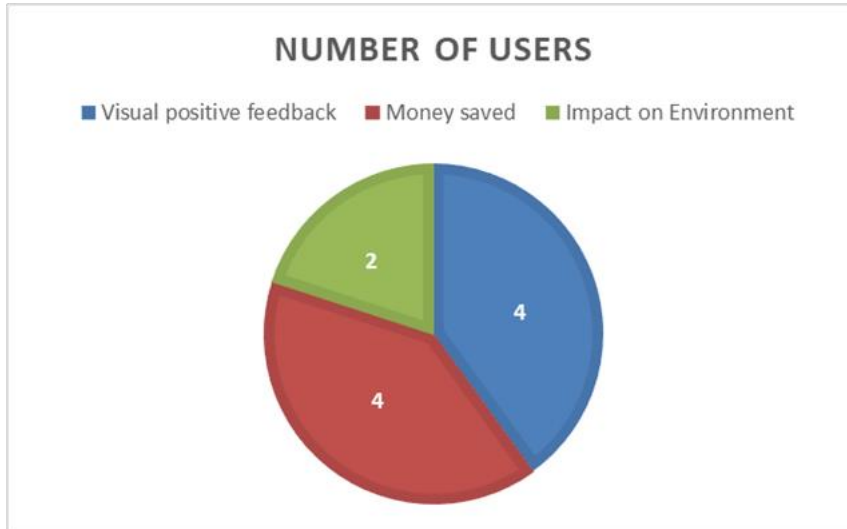
1. How often do you check your electrical usage?

<u>Frequency</u>	<u>Number of Users</u>
Only when bill arrives	6
Weekly	3
Daily	1
Never	0



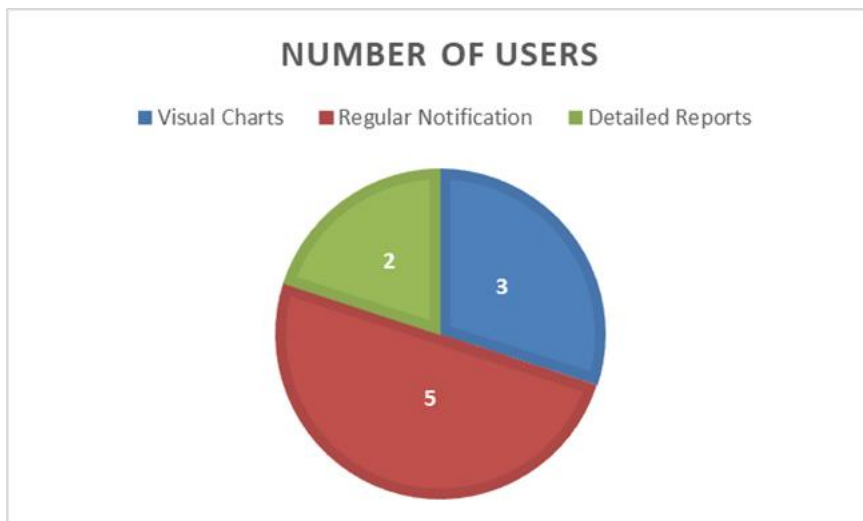
2. What makes you want to save energy?

<u>Frequency</u>	<u>Number of users</u>
Visual Positive feedback	4
Money saved	4
Impact on environment	2



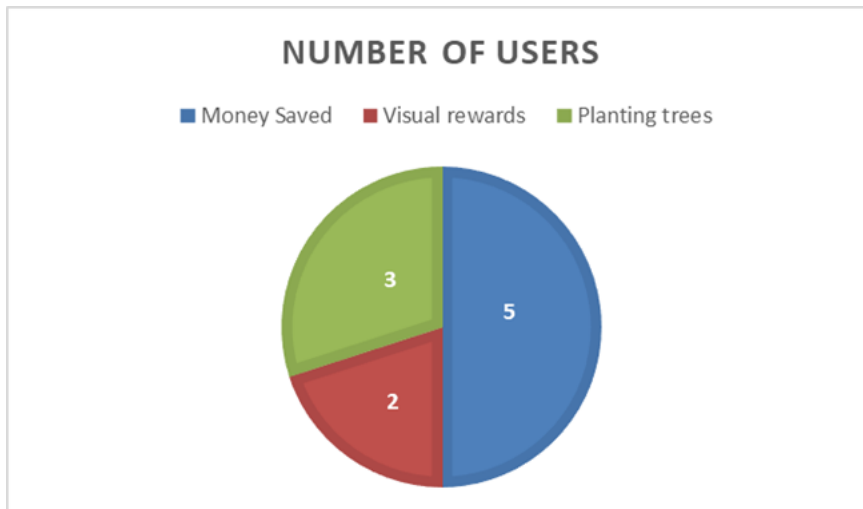
3. What's your preferred format for energy usage information?

<u>Frequency</u>	<u>Number of users</u>
Visual charts	3
Regular notification	5
Detailed reports	2



4. What reward system would motivate you the most to save energy?

<u>Frequency</u>	<u>Number of users</u>
Saving money on bills	5
Visual progress (e.g. Streaks)	2
Environmental Awards	3
(Real-time trees plantation)	



Insights gained:

It can be concluded from the survey that most users only worry about their energy usage when they receive the bills and do not engage much otherwise. Most of the users are mainly motivated by saving money while some are influenced by visual feedback or positive impact on the environment. Additionally, users prefer simple formats such as notifications or basic summaries rather than elaborate reports. This helps better understand the needs and adapt the design portion of the website accordingly.

Concise user needs:

Below are the precise and to the point user needs found so far by the conducted research:

- A simple and intuitive interface (website in this case) for tracking energy usage
- Clear and concise insights (e.g. percentage changes, comparisons over time)
- Visual feedback that highlights progress and improvements (e.g. Streaks system)
- Real-time incentives that contribute to environment sustainability (e.g. planting trees)
- Motivation through positive reinforcement (e.g. “you saved energy this week”)
- Easy comparison with past usage or average household benchmarks (based on a certain area point)

Define

This phase focuses on synthesising the findings from the empathise stage to clearly identify the core problem that the proposed solution aims to address. Based on the user research conducted, it is evident that while users are aware of the importance of managing their household energy consumption, they struggle to engage with existing methods due to their complexity and lack of clarity.

The survey results indicate that most users only check their energy usage when they receive their bills, demonstrating low levels of ongoing engagement. Additionally, users showed a strong preference for simple formats such as notifications and basic summaries over detailed reports or complex visualisations. This suggests that current solutions fail to align with user expectations for simplicity and ease of use.

Another key insight is that the primary motivation for users to save energy is financial, particularly reducing their utility bills. Secondary motivations include visual progress tracking and contributing to environmental sustainability. This highlights the importance of providing clear, immediate, and meaningful feedback to encourage behavioural change.

From these insights, it can be concluded that users are not lacking awareness, but rather **lack accessible, engaging, and easy-to-understand tools** that support consistent interaction and decision-making.

Based on the above analysis, the problem is defined as:

Users need a simple, intuitive, and engaging way to monitor and reduce their household energy usage because existing solutions are overly complex, not user-friendly, and fail to provide clear and motivating feedback for regular use.

This problem definition establishes the need for a solution that:

- Promotes regular engagement rather than occasional interaction
- Simplifies energy-related information into clear and concise insights
- Provides motivating feedback such as cost savings and visual progress
- Supports users with varying levels of technical knowledge through an accessible interface

By clearly defining this problem, a strong foundation is established for developing a user-centred solution that directly addresses the behaviours, motivations, and challenges identified during the empathise phase.

Ideate

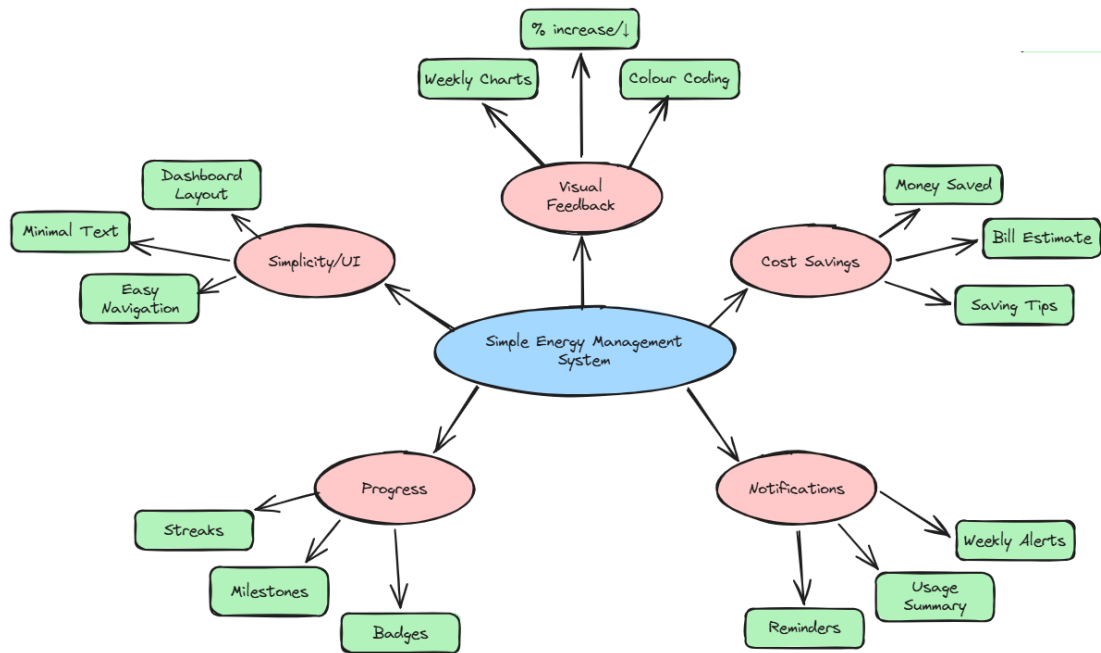
The ideation phase focuses on generating potential solutions that address the user needs identified during the empathise and define stages. Based on the research findings, the key challenge is to present household energy usage information in a way that is simple, engaging, and easy to understand without overwhelming the user. Since the survey results indicated that most users only check their energy usage when the bill arrives, the ideation process focused on features that encourage more regular interaction with the platform.

One idea developed during this stage was the use of simple visual summaries to present energy usage information. Instead of showing complex reports or large amounts of raw data, the website could display weekly or monthly usage trends through clear charts and percentage indicators. For example, users could easily see whether their energy consumption has increased or decreased compared to the previous week using colour-coded visuals. This approach aligns with the survey results where several users preferred visual charts and simplified summaries rather than detailed reports.

Another concept explored was providing motivational feedback related to cost savings. Since the survey indicated that saving money was the most common motivation for reducing energy usage, the platform could show users an estimated amount of money saved when their consumption decreases. For example, if a user reduces their energy usage compared to the previous billing period, the system could display a message such as “You saved approximately \$10 this month by reducing your electricity usage.” This type of feedback helps reinforce positive behaviour and encourages users to continue monitoring their consumption.

The ideation phase also considered introducing progress tracking features to keep users engaged. For instance, the website could implement a simple streak or milestone system that rewards users for consistently maintaining lower energy usage over time. Users could see visual progress indicators showing how many weeks they have stayed within a target usage range. This idea connects with survey responses where some users indicated that visual progress or environmental rewards could motivate them to save energy.

Finally, the design ideas focused on maintaining a simple and accessible interface so that users with limited technical knowledge can easily understand the information presented. A dashboard-style homepage could display the most important information, such as current energy usage, recent changes, and progress towards savings goals. By prioritising clarity and reducing unnecessary complexity, the platform can help users manage their household energy usage without feeling overwhelmed or confused.



Overall, the ideas generated during this phase aim to simplify energy monitoring while encouraging consistent engagement through visual feedback, cost-saving insights, and clear progress tracking.

Requirements

Functional Requirements

- F1.** The website must allow users to view their energy usage data in a simple format (charts or summaries).
- F2.** The system should provide weekly and monthly comparisons of energy usage to help users track changes over time.
- F3.** Users must receive triggered notifications when usage increases by more than 20% as compared to the last period.
- F4.** The website should provide simple tips and suggestions to help users reduce energy consumption.
- F5.** Users should be able to set goals (e.g. reduce energy by 10%) and track progress.
- F6.** The system must display estimated cost savings when users reduce energy usage.
- F7.** The website should include a reward or progress system (e.g. streaks or achievements) to motivate users.

F8. Users must be able to access the website on different devices (mobile, tablet, desktop).

Non-Functional Requirements

NF1. The website must have a simple and user-friendly interface to support users with low technical skills.

NF2. The system must have a quick response time (<3 seconds).

NF3. The design must be responsive and work well on mobile devices.

NF4. The website should maintain consistent layout, colours, and navigation across all pages.

NF5. The system must ensure data privacy and security for user information.

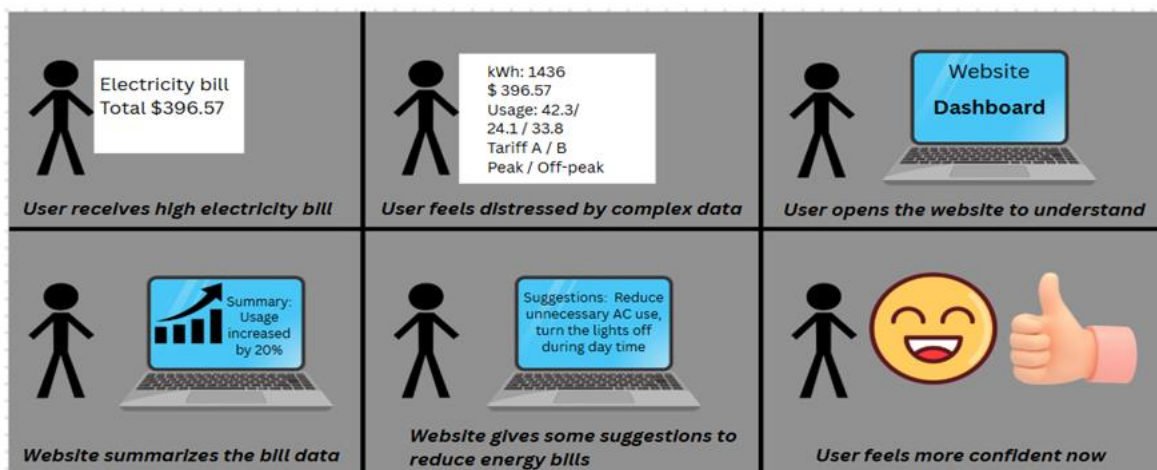
NF6. The website should be accessible, including readable fonts and clear contrast.

NF7. The system must be scalable for additional future features.

Storyboard

Storyboard 1

Scenario: A user receives a high electricity bill and is confused about the complex data and uses the website to better understand it.

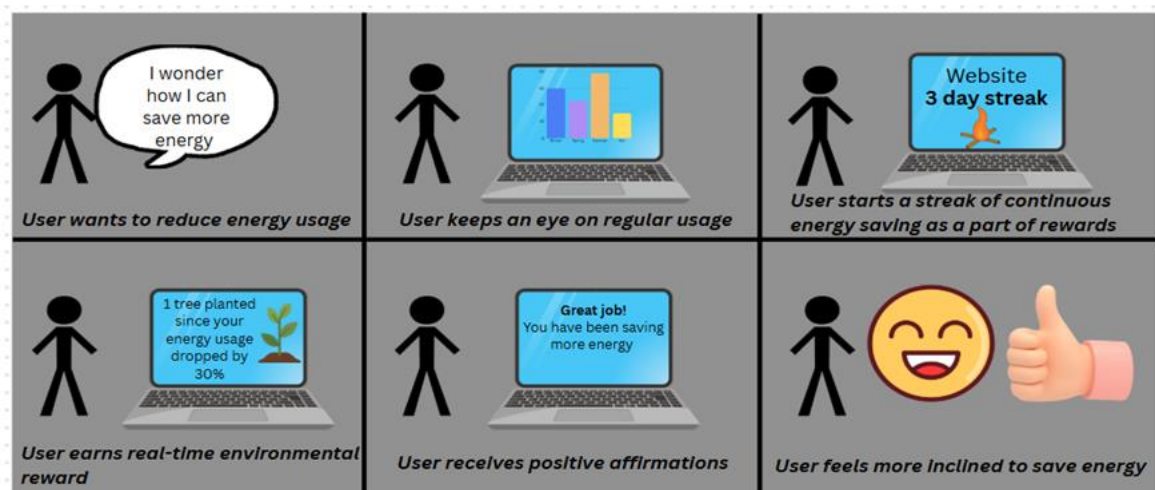


Description: At first, upon viewing the bill, the user feels quite overwhelmed and confused due to the complex information. To better understand the usage, the user looks up the website and clicks on the "Dashboard". The website provides a simple dashboard which shows the usage trends and changes via visual tools. On further clicking on the option of "Tips" the system also provides simple suggestions to reduce energy consumption. As a result, the user feels more confident in managing the bills now.

Connection to Design Thinking Methodology: This scenario depicts the part from empathise phase research, where users only check their bills upon arrival and find it hard to understand because of information overload. The design addresses this issue by simplifying the complex data into simpler terms, supporting suggestions to overcome it.

Storyboard 2

Scenario: This scenario represents how the user can feel motivated by the reward system such as streaks and environmental impacts.

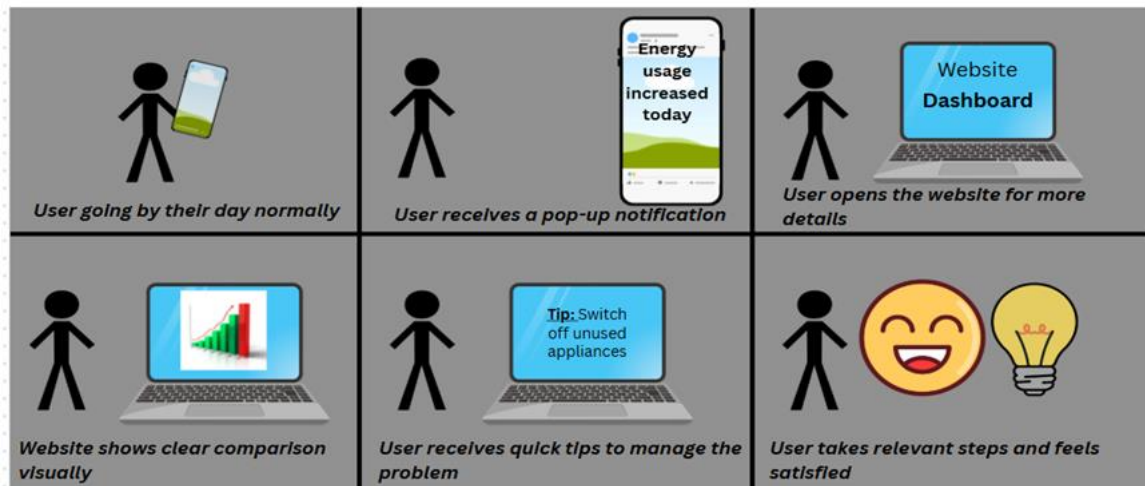


Description: The user engages with the website daily to monitor their energy consumption over time. As a result, the system introduces a kind of reward system called "Streaks". This acts as a fun way to encourage the user to keep maintaining low consumption. As this goes on, the user also gets positive feedback and some environmental awards such as tree planting. Such positive feedback keeps the user interested in continuing their progress.

Connection to Design Thinking Methodology: This storyboard links to the findings from empathise phase, where certain users expressed that they feel motivated by positive feedback and rewards. It also aligns with the Ideate phase where gamification elements like streaks and rewards were discussed to increase user engagement.

Storyboard 3

Scenario: This scenario shows when a user receives a pop-up notification and responds to it on the website.



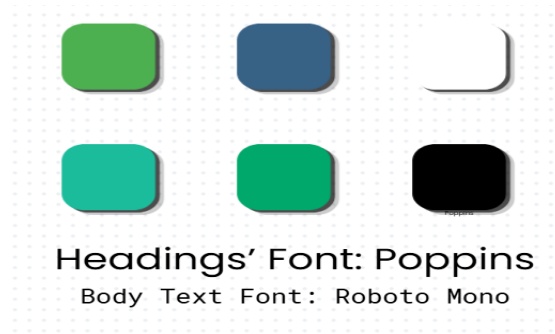
Description: While the user goes through their daily routine, the user receives a notification from the website that indicates higher energy usage. Then the user accesses the website's dashboard which provides a clear and simple comparison of current and previous usage. The system also provides some quick tips to mitigate it. Thus, the user can take actions accordingly to reduce the high usage of energy.

Connection to Design Thinking Methodology: This storyboard links to the idea of user preferences about notifications and updates in the empathise phase. This creates a strong commitment by providing easy-to-understand updates but doesn't overwhelm the user either.

Alternative Path: If the notification is ignored by the user, the system sends a reminder sometime later to keep the user in the loop.

Universal Design/Branding

The branding of the website aims to create a clear, modern, and user-friendly visual identity that matches the purpose of promoting energy awareness and sustainable behaviour. A consistent colour palette is used throughout the platform, with green representing sustainability and environmental responsibility, and blue representing clarity and reliability. These colours help strengthen the purpose of the website while creating a visually appealing design.



The design maintains consistency across all pages, including the dashboard, tips section, and feedback features. Elements such as navigation bars, buttons, and layout structures follow a uniform style, allowing users to easily recognize different sections of the website and navigate without confusion.

In addition, the website adopts a clean and minimal visual style to reduce cognitive load and improve usability. Simple typography, such as Poppins and Roboto and well-organized layouts, ensure that information is presented clearly, which is especially important for users with limited technical knowledge. Overall, the branding supports a professional and easy-to-use user experience while encouraging engagement with energy-saving behaviours.

STYLE GUIDE	
Primary Colour	Green (#4CAF50)
Secondary Colour	Blue (#386186)
Accent Colour	Jade green/Turquoise green (#1ABC9C / #00A86B)
Background Colour	White (#FFFFFF)
Headings' Size and Font	24 px and Poppins
Body text Size and Font	18 px and Roboto Mono

Brand Logo: As for the website's name and brand, we decided to go with something that seems personal to the user and is not too flashy.



Low Fidelity Prototype

Prototype Overview

The low fidelity prototype is a simple representation of the proposed energy management website. It focuses on the layout, navigation, and key functionalities without detailed visual design. The prototype was designed based on the user research findings from the empathise and define stages. It aims to provide a simple and easy-to-understand interface that helps users manage their energy usage without feeling overwhelmed.

The prototype includes four main pages:

- 1. Home Dashboard
- 2. Energy Usage Page
- 3. Tips Page
- 4. Goal Tracking Page

Prototype Description

Basic Navigation Flow: Users navigate the website via the Home Dashboard where they can access the rest of the options including energy usage page, tips page, streaks collection page and progress tracking page via a bar. They will also be able to navigate back to and through different pages.

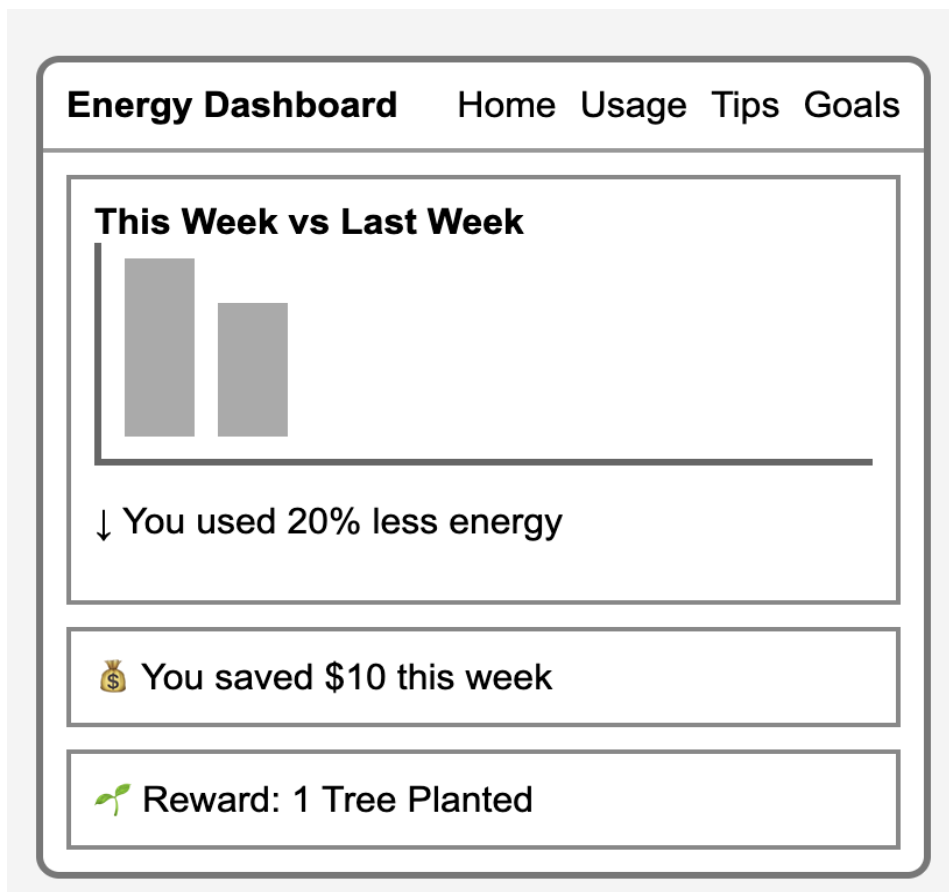
Home Dashboard

The Home Dashboard provides a quick overview of the user's energy usage. It displays a simple comparison between current and previous usage using basic charts.

It also shows:

- Estimated cost savings (e.g. "You saved \$10")
- Reward feedback (e.g. "Tree planted")

This design supports user motivation and avoids complex data presentation.

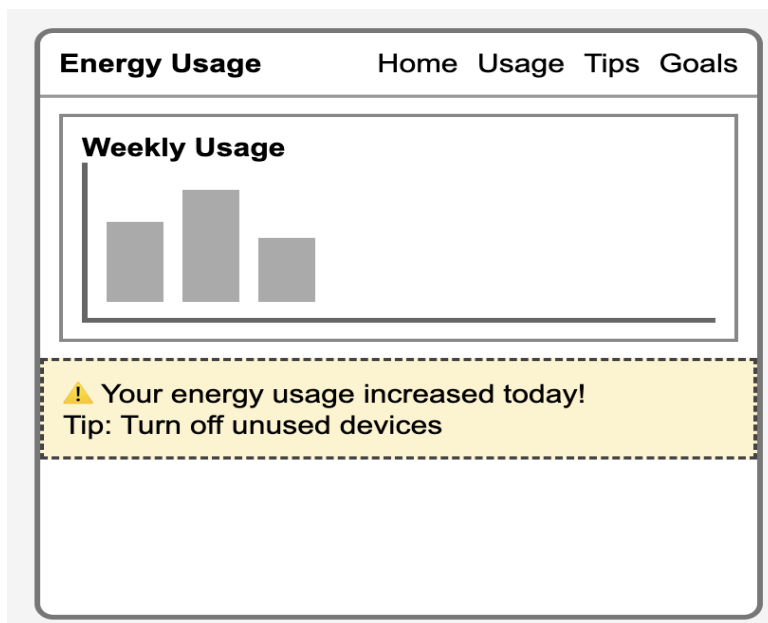


Energy Usage Page

This page presents energy usage data in a simplified visual format. Instead of complex reports, it uses basic charts to show weekly trends. A notification feature is included to alert users when energy usage increases. This helps users take immediate action.

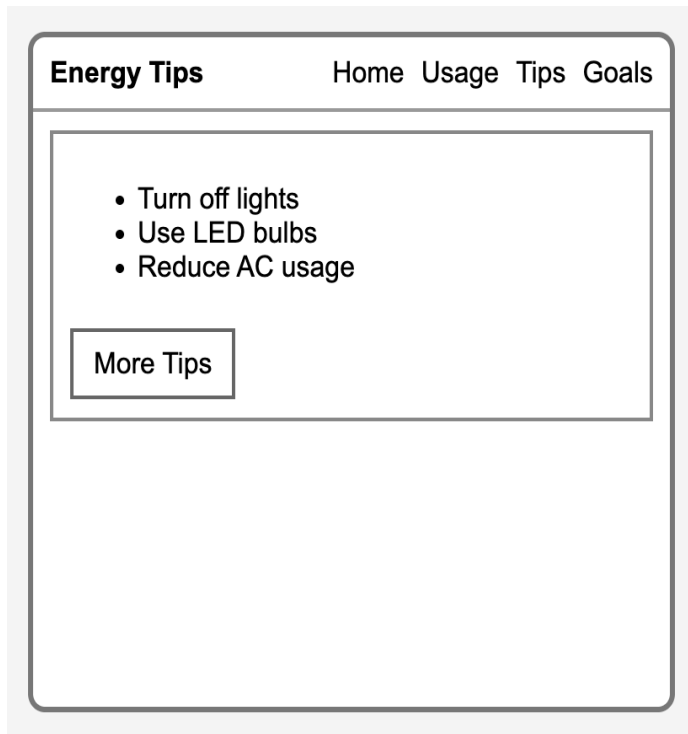
Example:

- “Your energy usage increased today”
- Suggested action (e.g. turn off unused devices)



Tips Page

The Tips Page provides simple and practical suggestions to reduce energy usage. The content is kept short and easy to understand to avoid overwhelming users. This supports users with low technical knowledge.

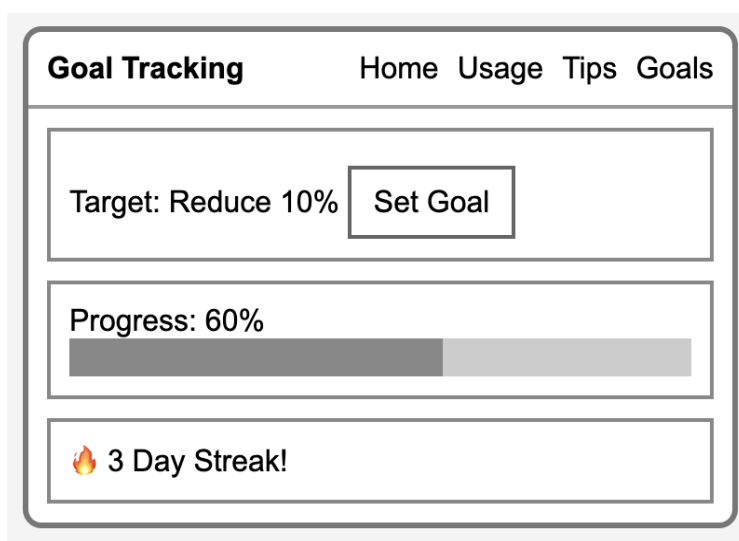


Goal Tracking Page

This page allows users to set energy-saving goals (e.g. reduce usage by 10%) and track their progress. It includes:

- Progress bar
- Percentage completion
- Reward elements such as streaks

This feature increases user engagement and motivation.



Design Rationale

The prototype design is based on key insights from user research:

- Users prefer simple and clear information rather than complex reports
- Most users are motivated by saving money
- Some users are encouraged by visual feedback and rewards
- Users do not regularly check energy usage unless prompted

Therefore, the prototype focuses on:

- Simple layout and minimal content
- Visual feedback (charts, progress)
- Cost-saving information
- Notification and reward system

Test Purpose

The purpose of the low fidelity prototype is to evaluate the usability and effectiveness of the design, and the results will be used to improve the design and usability of the assignment. Specifically, it aims to test:

- Whether users can understand energy data easily
- Whether the interface is simple and intuitive
- Whether users feel motivated by feedback and rewards
- Whether navigation between pages is clear

Evaluation Method

User Tasks

Users will be asked to complete the following tasks:

- Task 1: Find your weekly energy usage
- Task 2: Check if your usage increased or decreased
- Task 3: Find one tip to reduce energy
- Task 4: Set a goal to reduce energy usage

Data Collection

The following data will be collected:

- Time taken to complete each task
- Number of clicks
- Errors or confusion
- User feedback

Testing Methods

- **Observation:** Observation was used to understand how users interacted with the prototype in real time. While users were completing the tasks, their actions were monitored to identify any confusion, hesitation, or errors. This helped highlight usability issues such as difficulty in navigation or unclear features.
- **Interview:** After completing the tasks, short interviews were conducted to gather user feedback. Users were asked about their experience, including what they found easy to use, what they found confusing, and what they liked or disliked about the design. This provided deeper insights into user satisfaction and expectations.
- **Focus Group:** A small group discussion was conducted to explore user opinions and suggestions for improvement. Participants shared their thoughts on the overall design, features, and usability of the prototype. This helped identify common patterns in feedback and provided ideas for refining the design further.

Testing Results and Findings

The prototype was tested with a small group of 3–5 users in an informal setting to evaluate its usability and effectiveness. The aim was to understand whether users could easily navigate the interface, complete key tasks, and understand the information presented.

Users were asked to complete tasks such as finding their weekly energy usage, checking whether their usage had increased or decreased, locating a tip to reduce energy consumption, and setting a goal.

Observation was used to monitor how users interacted with the prototype. Most users were able to complete the tasks without major difficulty, which suggests that the overall layout and navigation are clear. Users were able to move between pages such as the dashboard, usage page, and tips section without confusion.

However, some minor usability issues were identified during observation. A few users took longer to locate the comparison between current and previous energy usage, indicating that this feature could be made more visible. In some cases, users hesitated before selecting options, suggesting that clearer headings or labels would improve usability.

After completing the tasks, short interviews were conducted to gather user feedback. Most users mentioned that they found the design simple and easy to understand. The use of basic charts and short descriptions helped them quickly understand their energy usage without feeling overwhelmed. This supports earlier findings from the empathise stage, where users preferred simple and clear formats.

A small group discussion (focus group) was also carried out to explore user opinions further. Users agreed that the reward features, such as streaks and environmental feedback, made the experience more engaging. However, some suggested that these features could be made more visually noticeable to increase motivation.

Overall, the testing results indicate that the prototype is effective in providing a simple and user-friendly experience. While the design works well, the feedback highlights areas where clarity and engagement can be improved.

Meetings

Meeting Location	Date	Length of Meeting	Attendees	Topics
N16 Lounge Area	12/03/26	40 Minutes	Nhan, Zoe, Khushi, Gia, Maide	• Introducing ourselves to each other. • Talking about the Assignment 1 Tasks
N53 Collaboration Area	19/03/26	23 Minutes	Nhan, Khushi, Ishan	• Assigning tasks to each member
N53 Collaboration Area	02/04/26	30 Minutes	Nhan, Gia, Maide	• Sorting out the remaining tasks
Online Teams meeting	14/05/26	30 minutes	Khushi, Ishan, Maide, Gia, Nhan	• Checking in how everything's going and minor editing

Team Contributions

Team Member	Contributed (Out of 100% in total)	Sections worked on by each member
Nhan	20%	Ideate and tweaked some other sections. Also did the final review of the entire document and recording the meeting
Ishan	20%	Has done the define section in the use of design thinking section, as well as reworked on the test purposes
Khushi	20%	Empathise, storyboards; and tweaked website logo, colour pallets and style guide. Did some final edits to some sections
Gia	20%	Worked on the requirements section, as well as the most of the Low Fidelity Prototype, with everyone else revising some sections
Maide	20%	Worked on the Introduction, as well as the Universal design/branding.
Zoe	0%	N/A: They have never been seen since week 2 so we've assumed they have dropped the class